

Sven Lüpke

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🔗 sven-luepke.github.io in sven-luepke 🌐 sven-luepke



Experience

Machine Learning Engineer, Delicious Data

- Led development of an internal transformer-based forecasting model from R&D to production, significantly improving accuracy on business-critical days and eliminating manual intervention
- Initiated and led the integration of tabular foundation models into our production forecasting pipeline.

since Mar 2025
Munich, Germany

Machine Learning Engineer - Working Student, Delicious Data

- Research and development on time-series forecasting and demand estimation
- Technologies: Python, Pandas, PyTorch, MLflow, Azure AI, Deep Learning

Oct 2023 - Mar 2025
Munich, Germany

Graphics Software Engineer - Working Student, Infineon Technologies

- Developed a calibration tool for an automotive head-up display
- Technologies: C++, OpenCV, Qt, CMake

Apr 2023 - Oct 2023
Munich, Germany

Research Assistant, Technical University of Munich

- Developed a real-time path tracer and analyzed a neural denoiser at the chair for computer graphics and visualization
- Technologies: C++, Vulkan, Ray Tracing, Python, PyTorch, CUDA

Oct 2021 - Sep 2022
Munich, Germany

Software Developer - Working Student, Rubinstein & Schmiedel

- Built a real-time market data management system for AI-driven trading systems
- Technologies: Python, Redis, PostgreSQL, REST, WebSocket, Docker

Jan 2020 - Sep 2021
Munich, Germany

Tutor: Practical Course Real-Time Graphics, Technical University of Munich

- Taught students the fundamentals of 3D computer graphics programming
- Technologies: C++, DirectX 11, HLSL

Apr 2019 - Sep 2019
Munich, Germany

Education

M.Sc. Informatics, Technical University of Munich

- Specialization: Machine Learning - **Passed with Distinction**

Apr 2021 - Jul 2025

B.Sc. Informatics: Games Engineering, Technical University of Munich

- Thesis: Injection of Physically Based Volumetric Effects into The Witcher 3

Oct 2017 - Mar 2021

Publications

Physics-Informed Latent Diffusion for Multimodal Brain MRI Synthesis Sven Lüpke, Yousef Yaganeh, Ehsan Adeli, Nassir Navab, Azade Farshad

Oct 2024
arxiv.org/abs/2409.13532 

- Presented at the [5th International Workshop on Multiscale Multimodal Medical Imaging](#) 
- Technologies: Python, PyTorch, Generative Models, Denoising Diffusion, Medical Imaging

Selected Projects

Generative Modeling for Inverse Molecular Design

- Interdisciplinary project at the chair for data-driven materials modeling at TUM
- Technologies: Python, PyTorch, Graph Neural Networks, Probabilistic Models

Oct 2023 - Jun 2024

[github.com](#) 

BlitzFX

- Graphics effect injector for The Witcher 3 downloaded by over 30,000 users
- Technologies: C++, HLSL, DirectX 11, Real-Time Rendering, Reverse Engineering

Nov 2021 - Mar 2023

[nexusmods.com](#) 

Roman Colosseum Reconstruction

- 3D model of Roman Colosseum inspired by old paintings and movies
- Technologies: Blender

Nov 2016 - Jul 2017

[cgtrader.com](#) 